



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KANNAPPAN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

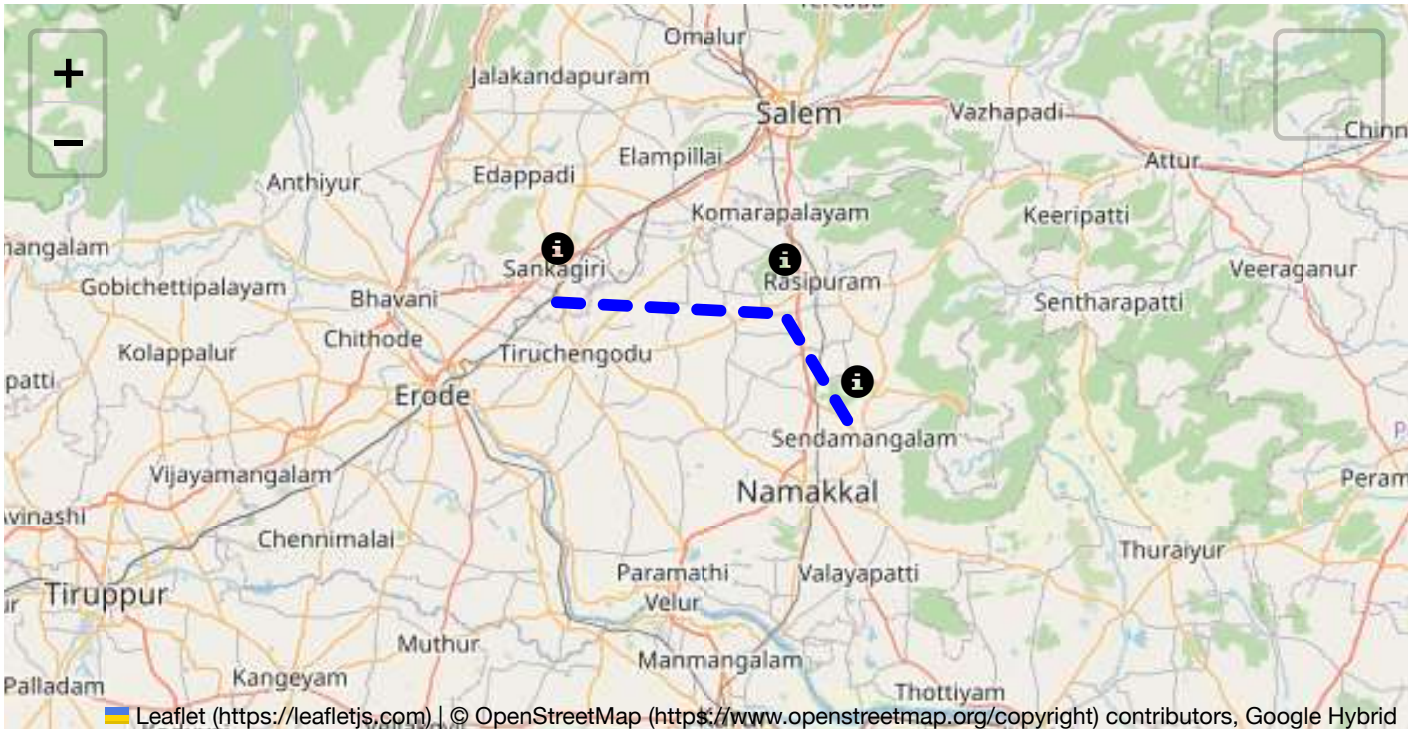
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

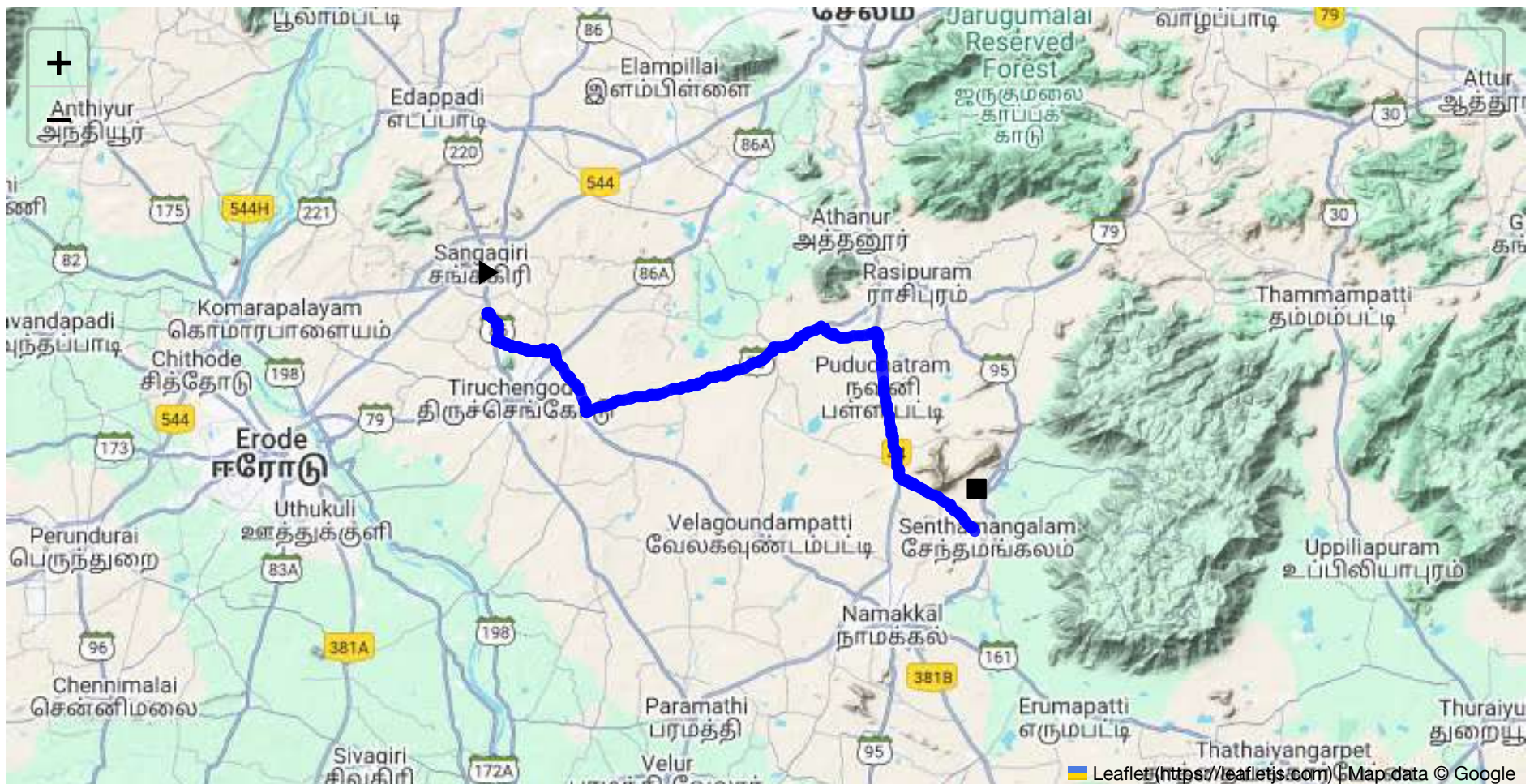
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 57.85 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours

Start: (11.4381, 77.8734)
End: (11.28446, 78.22614)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

The route starts from Sangagiri, Tamil Nadu, passing through Rasipuram and ends in Pachudayampatti, Tamil Nadu. The journey of approximately 57.85 km covers a blend of NH roads and local highways. This typically includes NH79 and other state roads, conducive to heavy vehicular traffic.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** Tamil Nadu generally experiences a tropical climate. It involves hot summers from March to May, monsoon rains from June to September, and mild winters from November to February.
- **Weather Hazards:** Monsoon rains can lead to waterlogging and reduced visibility. Flooding and slippery roads increase accident risks during the monsoon period.

3. Analysis of Traffic Patterns

- **Peak Hours:** Major traffic congestion can occur during morning hours (8 AM - 10 AM) and evening hours (5 PM - 8 PM).
- **Congestion-prone Areas:** Potential bottlenecks at intersections near Rasipuram and any toll booths along NH79.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** National highways like NH79 are generally in good condition, while state roads might have varying quality, with potential potholes and narrower lanes.
- **Infrastructure:** Adequate signage is present; however, road maintenance can be inconsistent, posing challenges for heavy vehicles.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** If NH79 is obstructed, consider using SH95 via local detours, though these roads may not support heavy vehicles as efficiently.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Transport Regulations:** Compliance with Tamil Nadu's stringent regulations regarding hazardous materials is crucial, requiring specific licenses and adherence to safety protocols.
- **Time Restrictions:** Transport of hazardous materials may be restricted during peak city travel hours or through sensitive zones.

7. Overview of Historical Incidents

- **Incidents:** Historical data indicates occasional incidents involving heavy vehicles, particularly during the monsoon season due to road conditions. NH79 has reported mishaps due to speed and visibility factors.

8. Environmental Considerations and Sensitive Areas

- **Environmental Considerations:** Ensure routes avoid ecologically sensitive areas. Follow protocols for chemical spill containment.
- **Sensitive Areas:** Proximity to agricultural lands in Rasipuram requires careful monitoring.

9. Analysis of Communication Coverage

- **Coverage:** Major routes have adequate mobile network coverage. However, rural sections near Pachudayampatti might experience intermittent communication lapses.

10. Estimated Emergency Response Times

- **Response Times:** In urban areas like Rasipuram, response times can be 20-30 minutes. In more rural parts, expect longer, at 40-60 minutes.

12. Overall Summary of Risk Assessment

- **Risk Assessment:** The route presents moderate risk due to varying road conditions and weather impacts. Proper adherence to regulations, regular vehicle checks, and monitoring weather forecasts are essential. Emergency plans should be ready due to potential delays in response times, and drivers should be trained in handling hazardous materials safely.

Overall, with careful planning and adherence to safety measures, the transport of hazardous materials along this route can be conducted with minimized risk. Continuous updates on road conditions, weather forecasts, and communication channels will aid in ensuring smooth transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
3	Turn	High	11.41981, 77.88078	15 KM/Hr
4	Turn	Medium	11.41876, 77.88487	30 KM/Hr
5	Turn	Medium	11.41829, 77.88568	30 KM/Hr
6	Turn	High	11.41859, 77.88692	15 KM/Hr
7	Turn	Medium	11.41761, 77.88739	30 KM/Hr
8	Turn	Medium	11.41716, 77.88935	30 KM/Hr
9	Turn	Medium	11.41630, 77.89077	30 KM/Hr
10	Turn	Medium	11.41603, 77.89479	30 KM/Hr
11	Turn	Medium	11.41439, 77.90025	30 KM/Hr
12	Turn	High	11.41377, 77.90073	15 KM/Hr
13	Turn	Medium	11.41375, 77.90128	30 KM/Hr
14	Turn	Medium	11.41389, 77.90161	30 KM/Hr
15	Turn	Medium	11.41325, 77.90313	30 KM/Hr
16	Turn	Medium	11.41276, 77.90349	30 KM/Hr
17	Turn	High	11.41230, 77.90485	15 KM/Hr
18	Turn	Medium	11.41262, 77.90506	30 KM/Hr
19	Turn	Medium	11.41273, 77.90524	30 KM/Hr
20	Turn	High	11.41329, 77.90978	15 KM/Hr
21	Turn	High	11.41424, 77.91433	15 KM/Hr
22	Turn	Medium	11.41409, 77.91458	30 KM/Hr
23	Turn	High	11.41208, 77.91551	15 KM/Hr
24	Turn	Medium	11.41218, 77.91651	30 KM/Hr
25	Turn	High	11.41347, 77.92096	15 KM/Hr
26	Turn	Medium	11.41319, 77.92144	30 KM/Hr
27	Turn	Medium	11.40833, 77.92362	30 KM/Hr
28	Turn	High	11.40566, 77.92355	15 KM/Hr
29	Turn	High	11.40557, 77.92343	15 KM/Hr
30	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
31	Turn	Medium	11.41332, 78.08072	30 KM/Hr
32	Turn	High	11.41463, 78.08067	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
33	Turn	High	11.42964, 78.11587	15 KM/Hr
34	Turn	Medium	11.42450, 78.15278	30 KM/Hr
35	Blind Spot	Blind Spot	11.42631, 78.15605	10 KM/Hr
36	Turn	High	11.32214, 78.17258	15 KM/Hr
37	Turn	Medium	11.28793, 78.22365	30 KM/Hr
38	Turn	Medium	11.28780, 78.22452	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
6	hospital	Government Hospital	11.3900664, 78.1606968	30 km/h	Medium
7	police	Police station	11.3844885, 78.1622883	30 km/h	Medium
8	police	Puduchathiram Police station	11.3845086, 78.1624257	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
2	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
3	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
4	college	Gnanamani Institutions	11.4210938, 78.157164	30 km/h	Medium
5	school	Paavai Group of schools	11.3994676, 78.1594685	30 km/h	Medium
9	school	RGR CBSE School	11.3790105, 78.1615844	30 km/h	Medium
10	school	RGR Matric Higher Secondary School	11.3717008, 78.1643097	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41981, 77.88078



Risk Type: Turn
Risk Level: Medium

Speed Limit: 30 KM/Hr
Coordinates: 11.41876, 77.88487



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41829, 77.88568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41859, 77.88692



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41761, 77.88739



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41716, 77.88935



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41630, 77.89077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41603, 77.89479



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41439, 77.90025



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41377, 77.90073



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41375, 77.90128



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41389, 77.90161



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41325, 77.90313



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41276, 77.90349



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41230, 77.90485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41262, 77.90506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41273, 77.90524



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41329, 77.90978



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41424, 77.91433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41208, 77.91551



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41218, 77.91651



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41347, 77.92096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41319, 77.92144



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40833, 77.92362



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40566, 77.92355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40557, 77.92343



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41332, 78.08072



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41463, 78.08067



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42450, 78.15278



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.42631, 78.15605



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.32214, 78.17258



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.28793, 78.22365



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.28780, 78.22452

